

Minor Project Presentation



CODEVEDA

API Code Integration for NAMASTE Portal & ICD-11 TM2

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1. Introduction

1.1 Overview

- This project focuses on integrating **AYUSH (NAMASTE)** and **ICD-11 TM2** into EMR systems.
- It uses **FHIR-based API** for healthcare data interoperability
- Acts as a **bridge between traditional and modern medicine systems.**
- Ensures standardized and structured medical records.

1.2 Purpose

- To enable **dual coding (AYUSH + biomedical diagnosis).**
- To improve **data sharing between healthcare systems.**
- To support **government digital health initiatives (ABDM).**
- To enhance **clinical research and decision-making.**

2. Literature Review

➤ This section reviews the existing systems and solutions related to the problem.

Sr. No.	Name of Solution/System	Features	Limitations/ Drawbacks
1.	NAMASTE Portal	Provides AYUSH morbidity codes, standardized terminology database	No integration with ICD-11, limited API support
2.	WHO ICD-11 API	Global disease classification, includes TM2 for traditional medicine	Does not support NAMASTE codes, lacks dual coding
3.	HAPI FHIR Server	Supports FHIR standard, enables healthcare data exchange	No built-in support for NAMASTE or ABHA authentication
4.	Ayushman Bharat Digital Mission (ABDM)	National digital health framework, supports EMR/EHR systems	Focus mainly on biomedical data, limited AYUSH integration



3. Problem Statement

- AYUSH (NAMASTE) and modern medicine (ICD-11) systems are **not integrated**.
- EMR systems mostly support **only biomedical coding**.
- This creates a **gap in patient data and communication**.
- There is no **common platform for dual coding**.
- It affects **data sharing, research, and insurance processes**.
- So, a **standard and secure integration solution is required**.



4. Proposed Solution

- Develop a **FHIR-compliant API microservice**.
- Integrate **NAMASTE + ICD-11 TM2 codes**.
- Provide **REST APIs for search and mapping**.
- Ensure **secure access using OAuth 2.0 (ABHA)**.

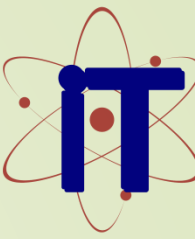
Innovativeness of the Solution

- First system enabling **real-time dual coding**.
- **Smart mapping engine** for automatic suggestions.
- Lightweight and **scalable microservice architecture**.
- Future-ready with **AI/ML integration**.



5. Objectives

- Develop a **FHIR-based API** for integration
- Enable **dual coding (NAMASTE + ICD-11)**
- Ensure **secure data access using ABHA (OAuth 2.0)**
- Improve **interoperability between systems**
- Support **accurate and standardized health records**



6. Key Features/Benefits

- **Auto-complete search** for medical codes
- **Smart mapping** between NAMASTE and ICD-11
- **FHIR bundle support** for easy EMR integration
- **Secure authentication & audit tracking**
- **Reduces manual work and errors**
- **Improves data accuracy and efficiency**

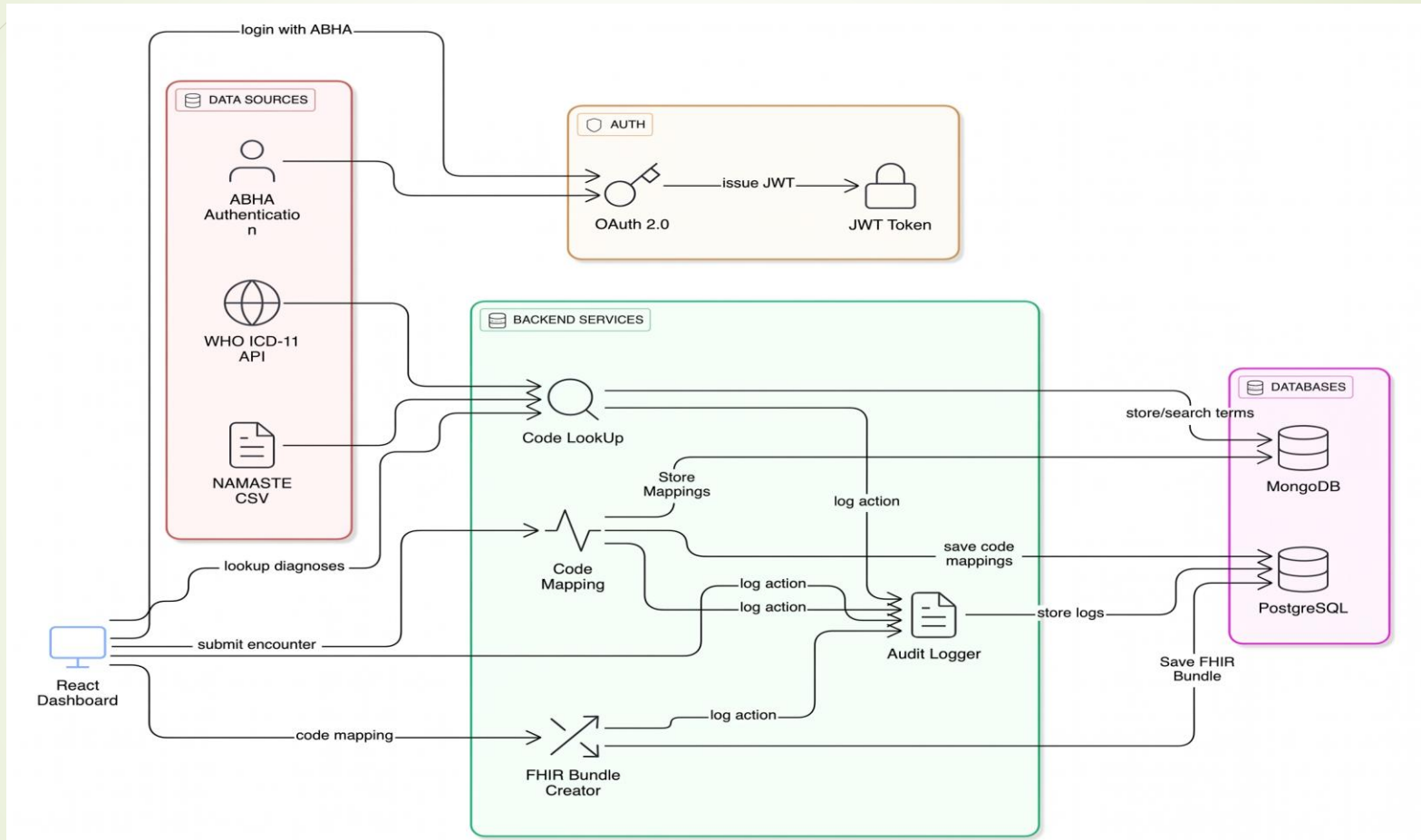


7. Society Impact

- Better **integration of traditional and modern healthcare**
- Improves **patient record management**
- Helps in **insurance claim processing**
- Supports **government health initiatives (ABDM)**
- Promotes **growth of AYUSH sector**
- Enables **better research and data analysis**



8. Block Diagram





9. Utility/Applications

- Used in **hospitals and clinics** for better EMR management
- Helps **doctors** to record both AYUSH and modern diagnoses
- Useful for **government health programs (ABDM)**
- Supports **insurance companies** in claim processing
- Helps in **medical research and data analysis**
- Can be used by **EMR/EHR software developers**
- Improves **data sharing between healthcare systems**
- Useful for **AYUSH practitioners** for digital record keeping



REFERENCES

- **ResearchGate Paper**
https://www.researchgate.net/publication/389801966_NAMASTE_portal_data_analysis_and_understanding_of_Ayurveda_medical_records_An_initial_evaluation
- **Indian EHR Standards (2016)**
https://www.nrce.in/download/files/pdf/nrces_ehr_stand_india.pdf
.
- **ICD / NAMASTE Portal**
<https://namstp.ayush.gov.in/#/index>
- **PIB Government Report**
<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2000638>
.
- **PubMed Research Article**
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10410508/>
.



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